WorldMirror: Place-Based Exploration as a Tool for Personal Reflection

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***Abstract*—People are often limited to their own perspectives, which are deeply shaped by their surroundings. Place – that is, the geographic location in which people live and interact with one another – is enriched by historical, cultural, and social contexts which subsequently influences how people think and behave. People often perceive locations, especially unfamiliar ones, as inherently distinct from the self, merely as places to visit and learn about rather than as mechanisms for personal development. Yet, people's internal worlds are shaped by the same forces which shape surroundings. Individual habits, beliefs, values, and attitudes evolve through exposure to societal, cultural, and environmental pressures. Whether consciously or subconsciously, people make efforts to improve self-concept and overcome personal challenges through reflection, self-awareness, and seeing issues from a new perspective. Oftentimes, it is the last element that enables the first two to occur. There is an opportunity to reconcile people's internal worlds with their external world, particularly through providing people with contexts that are disparate from their own. In this manner, we welcome alternative perspectives, grounded by physical locations, to help individuals reveal current thought patterns, make comparisons, and ultimately gain personal insights. In this paper, we present *WorldMirror*, a system that uses a mixture of LLM discussions and interaction with videos to help users gain a deeper understanding of their problem, learn different ways to approach their problem, and learn about the world. Given a user problem, the LLM prompts the user to learn more about their problem and connect it with different aspects of the user's life to identify underlying root causes. Then the LLM provides locations around the world that can teach the user new ways of thinking about their problem. Through interacting with a video of people living their daily lives at the location and further discussion with the LLM, the system pushes users to learn more about their problem and the world. A preliminary study with one participant using the system suggests that WorldMirror can understand the user's problem thoroughly and identify root causes but struggles to help users grow in their own thinking about their problem and the world.**

I. Introduction

Many people seek to better understand themselves, overcome personal challenges, and make sense of their place in the world. However, the ways individuals interpret their experiences are often constrained by the cultural assumptions, social norms, and perspectives that surround them. When people repeatedly approach a problem through the same framework, it can become difficult to imagine alternative ways of understanding themselves or their circumstances. Other cultures and ways of life could offer a different angle. Every place reflects a set of values that people there have built their lives around. Seeing how someone else has organized their life around the same kind of need one is struggling with can shift how one sees their own situation. However, learning about other cultures and ways of life is often limited to surface-level exposure. People frequently encounter other parts of the world as distant, exotic, or abstract cultural phenomena rather than as lived experiences shaped by real human values, struggles, and aspirations. Without meaningful engagement with the perspectives of others, it is difficult to develop the empathy and understanding necessary to challenge one's own assumptions. As a result, individuals may struggle both to connect deeply with people from different backgrounds and to recognize alternative perspectives that could help them navigate their own lives. There is a need for experiences that foster genuine human connection across cultures while enabling reflection on the assumptions that shape how people understand themselves and the world around them.

Existing place-based content severely reduces the dimensionality of locations. In reality, places are complex, interacting, and ever-evolving systems shaped by geography and the environment, culture and society, economics, and history. Rather than capturing all of these facets, current solutions represent places as a collection of highlights. This creates the opportunity to reintroduce this complexity and properly represent places as full-fledged models of life organization. At the same time, generic overviews about a place also sacrifice what places can mean to users as these are based on assumptions for what users should know versus what users are actually curious about. In turn, user engagement and relevance both become severely reduced. Existing solutions also limit user exploration, providing information answering the user's questions. While efficient in practice, curiosities often emerge naturally by interacting with the place itself. There is an opportunity to shift away from requiring users to have predefined interests to letting those interests unfold without prior conception. Collectively, these current obstacles make it difficult to find meaning and personal connection with unfamiliar places, even though connection can be formed at a distance. While familiarity and exposure is arguably the fastest vehicle for connection, connection is also enabled by shared values and sentiments as well as emotion. Places carry meanings which justify why certain things may exist in that place. It is hard to decipher meaning from observation alone, which emphasizes the need for platforms that push observation into critical thinking – understanding and reflection. [Design Name] seeks to make place-based learning a more active process, relying simultaneously on external observation and internal reflection, to make such content more accessible and memorable.

Our solution achieves two complementary purposes – (1) world learning and (2) self-reflection. In WorldMirror, one cannot be achieved without the other, which reduces the need to rely on multiple tools to achieve both outcomes simultaneously.

Existing solutions separate information, exploration, and self-reflection. By combining them, this design creates a novel, personalized experience where users can learn about a place while uncovering insights from that place that are directly applicable to their own lives. Ultimately, the user gains greater self-awareness, exposure to new perspectives, and leaves with next steps that help address their need or misalignment. The system begins with a diagnostic in which users can explain a personal problem or misalignment they are experiencing in their own lives. Because these can often be hard to articulate, our solution employs a chat bot to help clarify this tension through asking further questions and providing users with sets of options to select amongst depending on what the user feels best captures their problem. From there, the system generates a list of locations which, based on the user's latent needs, can plausibly push users to think about their struggle differently. After allowing the user to choose one of the recommended places, the system progresses to a virtual walkthrough. As the user walks through this digital simulation, they can take notice of specific features in the environment, whether that be objects, people, or activities. From there, users can choose to learn more about these entities by clicking into them, triggering a conversational chatbot. The chatbot initially collects user's initial impressions of the entities they clicked on and through this dialogue, reveals information and context about the entity's relevance or relationship to the place. This context is revealed in piecemeal and in through various media types to support user discovery and evidence-collecting. As the user wraps up their immersive experience in the location, the system offers a summary of external and internal learning arcs. This two-fold summary cleanly ties in the information about the place explored along with personal patterns uncovered through the interaction. It prescribes users with some recommendations or next steps which they can directly apply to their own lives, closing the loop with actionable insight. The crux behind this summary is that it distills knowledge gained from a place to the values of that place which users resonate with and show how those values can be transposed from that place to their everyday life.

Our design is based on the principle that places are not just destinations, but models of life that can be used as mirrors for the self. Every place reflects a set of underlying values about community, time, space, labor, and belonging that its inhabitants have, consciously or not, organized their lives around. When a person observes a place not as a tourist but as an investigator of their own situation, those values become legible in the objects and scenes around them. And this enables them to start comparison, reinterpretation, and growth.

The core idea of our design therefore is juxtaposition: we present an unfamiliar place to a user who has articulated a personal problem and misalignment, and we present an encounter between what they observe externally and what they are grappling with internally. The place helps the user reframe the problem. By surfacing the ways another culture has organized life around the same underlying needs like connection, purpose, belonging, users are invited to recognize that their own current way of living is not the only possible one. This distance from one's default framework is what creates the opening for reflection.

This juxtaposition is personalized at both ends. On the world side, the system selects locations based on the user's specific problems rather than generic appeal. On the self side, the user drives the discovery as their curiosity determines which objects they click, which questions they ask, which observations they carry into the reflective conversation. The AI helps the user to arrive at change and insight. This design choice treats the user as the author of their own understanding rather than a recipient of it, which we believe is essential for any insight that is meant to last.

We conducted a user study with one participant. The user discussed a problem he faced with the AI chat and then picked a location to explore. Next, he made observations out loud while watching a walking tour video of the location and chatted with the AI afterwards with the observations inputted as context to shape the conversation. I observed his interaction focusing on 1) what he learned about the location, 2) what he discovered about his own problem, and 3) the effect of world learning on shaping his perspective towards his problem.

The results showed that the prototype had mixed results in achieving the goal of world learning and self-reflection. The user felt that the AI was able to accurately connect other unstated underlying problems and the problem he provided; however, he also stated that he already knew about the connection and thus did not feel the AI provided anything beyond reinforcement to his understanding of the problem. In terms of world learning, the user picked a location he was already familiar with. While he learned a new fact about the location, he treated it as trivia and did not gain a deeper understanding of how people live in the location compared to before the interaction. Finally, observations of the location and people living in the location did not guide him towards a different view of his problem. Instead, it reinforced the problem's existence, and he felt the AI kept pushing a perspective to solving the problem he did not find relevant.

This paper makes the following contributions: 1. A design space with AI-mediated virtual exploration of the place that elicits personal reflection 2. A system that combines AI inquiry into personal problems, culturally-grounded and personalized place recommendations, immersive street view exploration and the conversational AI that enables the users to connect their investigations of the place to themselves.

A. Related Work

Our work draws from two bodies of work: 1) Research on how people explore and interact with places; 2) How do people come to understand their personal values. Through examining these works, we identify the gap that they fail to address.

*1) **How people explore and interact with places**:* A consistent finding through our literature reviews is that places are not "passive backdrop" but instead a "dynamic entity that actively influences identity, belonging, and comprehension." The article "To Learn is to Belong: Experiential Learning and Indigenous Connection to Place" illustrates this through Dr. Jonathan Pitt who frequents the Diver Airfield, a place which holds great historical significance. Instead of simply walking through the space, Dr. Pitt engages with it through his senses, reflects on the experience, and assimilates novel insights into his understanding. Inspired by indigenous culture's connection to place (known as indigenous ways of knowing), the article argues that this affective engagement, this felt sense of a place should be treated as a way of knowing in itself, instead of a secondary to traditional way of receiving information. Our design seeks to replicate this multisensory experience digitally in a manner that still generates active reflection, meaning, and learning without feeling forced [1].

Gesthuizen, Tan, and Kidman's study of virtual worlds in sustainability education shows this is achievable. They find that immersive virtual environments can produce genuine curiosity and emotional engagement, not just information retention, when that immersion is scaffolded with guided questioning that pushes users to think and reflect. The key insight of this work is that the virtual environment alone does not do this; facilitation is what produces the shift. Their facilitation, however, is oriented entirely outward: students reflect on sustainability, on the world being built, on environmental systems. What the design does not do is turn that reflection inward. Our system makes this pivot as the AI's facilitation is directed not at the place itself but at the intersection of what the user observes and what they are experiencing internally [2].

Guntarik, Davies, and Innocent's TIMeR project makes a related point about how exploration is structured. Through an augmented reality audio-game, players move through a university campus uncovering alternate cartographies including histories, Indigenous knowledge, and ways of inhabiting space that are invisible on the surface. The finding is that discovery-oriented movement through a place makes it legible in ways that passive observation cannot: users come to see differently because they moved through it with intent. However, TIMeR only reveals deeper meanings about the place itself, and does not connect that discovery to anything about the user. Our design borrows the discovery-oriented structure but reorients what the discovery is for as when a user clicks into an object and the AI begins asking questions, the goal is to use what was found in the place as a prompt for the user to examine something in themselves [3].

It is proven that learning about places largely different from one's own can alter one's own values and perspective of their own life. In one study involving students in Caracas, Venezuela, and Albany, United States, the students were divided in groups and tasked with creating a video focused on popular expressions in both cultures. The projects were meant to capture aspects of everyday life (e.g. street art, performance spaces, contemporary media of social interaction) as opposed to general highlights. This framework, called Collaborative Online International Learning (COIL), is intended to evoke empathy, transcultural awareness, and equitable collaboration. The module provides participants with new information on expression and tradition from interacting with their teammates abroad, while encouraging participants to question and update their previously held assumptions. In the process, students naturally recognized similarities and differences which enabled them to find "novel terrain" in their own contexts and create what the article describes as a "third culture" built on collaboration. This is relevant to our design because it intends to transform worldviews. People are often limited to the perspectives belonging to themselves and those around them which tend to be similar. By giving users exposure to wholly different perspectives, as did the students in the study, we anticipate that users can approach their personal misalignments with a more rich and open perspective. Below the surface, users can break current thought patterns and realize new possibilities [4].

*2) **Self Reflection and Personal Values**:* The second body of work addresses how people can be made aware of their own values and start self-reflection. Pommeranz and Detweiler's work on self-reflection and value-sensitive design brought up a problem: most people do not have explicit access to their own values. They argue that eliciting values from stakeholders is an inherent difficulty as meanings, nuances, and interactions of values are complicated to express in a simple ranking of abstract values, and even a single person's opinion about the importance of a value can change based on their current context. Their proposed solution is that value elicitation tools need to situate reflection in real-life moments rather than asking people to reason about values abstractly, because without a real-life context, a situation in which a value serves as a guiding principle for a decision, or in which the violation of a value is apparent, elicited value profiles may be based on spontaneous thoughts and therefore biased. To address this, they develop what they call "situated values": values made legible not through questionnaires but through concrete encounters with specific moments that activate them. Their exploratory study uses photo elicitation which is asking participants to photograph what matters to them in everyday life and then talk through those images, and finds that this concrete, contextual approach surfaces values that abstract surveys miss entirely.

What this tells us is that the path to self-knowledge is not introspection in the abstract; it runs through particular objects, scenes, and moments that carry personal weight. This inspires our design's use of place-based visual exploration as the material for reflection. When our user walks through a virtual street and clicks on a bench, a market stall, or a group of people gathered outside, those objects help users access and articulate something that would otherwise stay implicit. The AI's conversational questioning then plays the role of the photo elicitation interview, drawing out what those observations mean to the user personally.

Pommeranz and Detweiler's framework relies on the user's own life as the material: you photograph your own world and reflect on it. Our design inverts this. We use an unfamiliar world as the material for reflection on one's own life, on the premise that encountering how other people have organized their values into a place can reveal, by contrast or resonance, something about the user's own values that their immediate environment cannot. The situated reflection is not about what already exists in the user's life but about what might be possible, a reframing enabled by the distance of another culture's way of living [5].

*3) **The Gap**:* Across these bodies of work, no existing system combines outward world learning with inward personal reflection in a single, user-driven interaction. Place-based learning tools surface meaning in place but do not connect it back to the user's life. Self-reflection tools work from the user's existing world and do not use unfamiliar cultural material as a mirror. Immersive learning environments produce engagement but keep the reflection oriented outward. COIL produces the cross-cultural encounter but requires institutional infrastructure and another human. WorldMirror presents a system in which exploring another way of living and examining one's own tension are not two separate activities but the same one.

II. Design Space

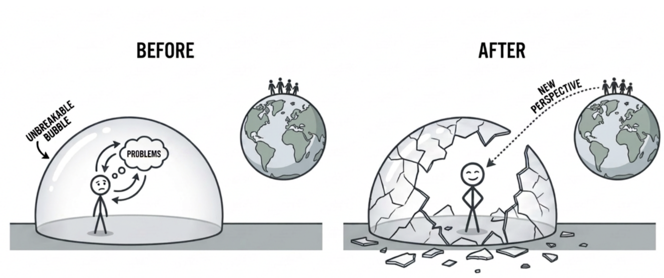


Fig. 1. Conceptual Representation of Design

A. Dimension 1: Reframing Places as Models of Life Organization Through Context

One dimension is the framing of places as not merely destinations, but rather as models to organize life. The idea behind this is that each place is a collection of physical environments, social norms, cultural conventions, and institutions that shape lifestyles. There is a reciprocal relationship here: (1) people shape the environments in which they live and (2) in turn, spaces shape the realized experiences of individuals. Existing place-based content primarily serves tourism or entertainment purposes, flattening places to their attractions and experiences. By shifting attention away from what exists to why they exist, we treat places as dynamic systems underpinned by historical, social, and environmental forces. The design challenge which surfaces is creating systems that encourage users to view places as alternative lifestyles/perspectives instead of merely physical settings.

To address this issue, we introduce context-backed observation. Users can interact with objects and people within the exploration interface to gain knowledge on the environment, histories, economics, and sociocultural factors that shaped them. Context can come in many mediums, from text, quotes, audio, and video, all of which supplement the main exploration interface. Prior spatial information research has found that integrating multimedia documents in 3D models greatly enhances the way a user can learn about and understand a territory [6]. The goal of provided context changes observation from a passive process to an active one. When someone passively views an object or a place, they take notice of its visible attributes without understanding the significance. The gap between seeing and understanding can be effectively bridged with contextual information. To illustrate, imagine a user exploring Copenhagen. They take notice of the city's high quantity of benches. Our design enables users to dive deeper. By clicking on a bench in this virtual representation of Copenhagen, users learn that benches and public spaces as a whole in Denmark are designed around the concept of hygge [7]– a quality of coziness and shared presence – reflecting the city's prioritization of human-centered urban planning. At its core, it reflects a greater longing for communal connection and well-being, emerging as a reaction to otherwise dark and cold Danish winters. Unless previously acquainted with a society's values and background, such knowledge is difficult to extrapolate from observation alone. As shown, our design advances the progression from object $\rightarrow$ observation $\rightarrow$ context $\rightarrow$ interpretation rather than short-circuiting at object $\rightarrow$ observation. Objects, people, and other features in the landscape become vectors to communicate deeper knowledge which may not otherwise be accessed. The use of multimedia to provide such context aims to improve user engagement, comprehension, and recollection. With this expanded flow, users now view places systematically – existence becomes rooted in reason and that reason is often unique to that place, embedding a layer of world learning.

B. Dimension 2: Learning through user-guided discovery and curiosity

While context is important, we must consider how that context is presented. Traditional learning materials about places present context explicitly, often as agglomerations of facts, images, and explanation. This format poses three major issues. For one, existing systems pre-determine what is relevant for users to learn about. Travel guides, for example, choose a subset of landmarks and attractions which the author may find noteworthy, but that the user has little to no interest in. The same is true for documentaries and encyclopedias, where content is filtered based on what authors believe people should know rather than extending flexibility to their audience in choosing what they would like to learn about. Different users may be drawn to different facets about a place: one user may be curious about architecture, another about nature, and another about food. Granted, users can refine their searches to cater their specific interests, but even so, users must know what they are interested in beforehand.

This segways into a second issue with current place-based educational tools, that is, users must be conscious of their interests in advance. While not inherently problematic, it misses the potential opportunity to evoke new and unexpected interests. For instance, a user may begin exploring a place like Alaska with an initial interest in phenomena like the Northern Lights and extreme weather but through exploring a simulation, that curiosity may evolve to how residents adapt their life to such environmental constraints (e.g. the impact of limited daylight hours on daily routines). This natural unanticipated curiosity may not manifest or be more difficult to achieve if users are pre-specifying topics. A third limitation is that existing tools, which often adopt a fact-explanation format, answer questions before users can even ask them. Users become inundated with information the moment they enter their query. While presenting information upfront can be more efficient – providing everything at the user's disposal – we miss a critical part of the learning process, which is inquiry. Learning through a series of successive questions as opposed to instant exposure is proven to be more effective. According to metacognitive theory, the reason is that discovery, this is due to the fact that individuals would have to stop and think about the significance of the information at hand while activating their own schema by combining new information with what they already know to generate questions [8]. Direct transmission of facts, on the other hand, does not encourage this personal discovery/exploration [9].

Our design employs a discovery-based approach to address each of these issues. To address the first two issues, existing solutions pre-filtering knowledge and users needing to enter with predefined interests, our design immerses users in a sandbox space presenting several points of interest which users then have the capability to choose which to interact with. In other words, our interface highlights a number of elements that users may notice, which are not limited to landmarks/attractions but also include everyday objects. The relevance of these objects are determined by the individual user. Users can choose which items to attend to rather than being forced to learn about what others may predefine as important. In this manner, each user chooses a unique subset of points, allowing for customized learning paths since different users may attend to different elements. Still, we initially present a diverse set of points of interest to allow users to sift through and identify what is meaningful to them and what is not.

To address the third issue where the use of direct instruction in current solutions stifles exploration, our design introduces questioning. When engaging with a point of interest, our design does not immediately administer context, instead that context must be probed through user inquiry. Users can ask questions and follow-up questions at their discretion, allowing information to be obtained through curiosity. Essentially, users uncover information/evidence, which makes the content more memorable (as users must deliberately as questions on their own) and more digestible [8].

C. Dimension 3: Questioning to stimulate personal reflection and connection

Existing place-based content externalizes locations. When learning about locations beyond your immediate surroundings, there is a physical separation between the place and self which makes it difficult to truly connect with that location unless the individual possesses prior knowledge or experience. While existing solutions explain how a place functions and what it means to the people living there, they rarely help users connect that information back to their own lives. As a result, the place and its civilians are isolated from the user. This constrains user ability to introspect and make connections across contexts (e.g. where the user considers home versus the location the user is learning about), which is needed for cognitive reinterpretation and ultimately make change. (source on what drives change)

Literature on place attachment, which is defined as how well a place meets an individual's functional needs as well as incorporates into sense of self [10], suggests that unfamiliarity does not automatically translate to meaninglessness and placelessness. In fact, unfamiliar locations can provide meaning if they are enabling, "aesthetically pleasing and are usable" [11]. Yet, even individuals living in the same physical place, do not experience the place in the same way as mental maps differ from person to person and from culture to culture [12]. By this logic, places can surface different meanings to different observers, making them a potentially powerful tool for personal introspection.

Our design seeks to foster connection by prompting users to interpret the values and behaviors they observe in an unfamiliar digitally-rendered place in relation to their own experiences. Objects/places/people serve as an entry point for guided questions which resemble journal-prompts (e.g. "why do you feel drawn to this object?", "what is something you observed that you wished existed in your own community?") as opposed to factual recall (e.g. "what is this object used for?"). By using the second-person, the design naturally feels conversational rather than one-sided. For instance, when exploring Hanoi, Vietnam on the interface, one may observe many people clustered around cafe tables, engaging in a wide range of activities from talking to studying. To provide context, cafe culture is quite popular in Vietnam where people can spend hours without pressure to leave and socialize. An appropriate reflection prompt may be "What places make you feel comfortable staying rather than simply passing through?"

Such journal-like questions make the user actively reflect in a way passive observation does not guarantee. Ultimately, the design intends to portray places as a mirror to oneself by transforming external observation into internal observation. Objects and other entities are anchors to ground personal introspection into something tangible.

III. System Description

A. User Interface

WorldMirror is designed for people who are faced with a problem they feel like they can't overcome, even after talking to those around them. The system combines information, immersion, and self-reflection to create a personalized experience of a location in the world for the user that lends perspective to their problem. On a higher level, users are able to talk with the system about a problem they have. Then, WorldMirror returns three locations and blurbs that provide insight to their problem. They then pick a location they are interested in and watch a couple minutes of a video moving through a local street to learn more about how locals live in the area. While watching the video, users note down what they are interested in which is then put into the AI chat. The AI chat then uses the user's interests to teach the user about the location as well as provide a new perspective on their problem. The inputs and flow of WorldMirror are shown in figure 1.

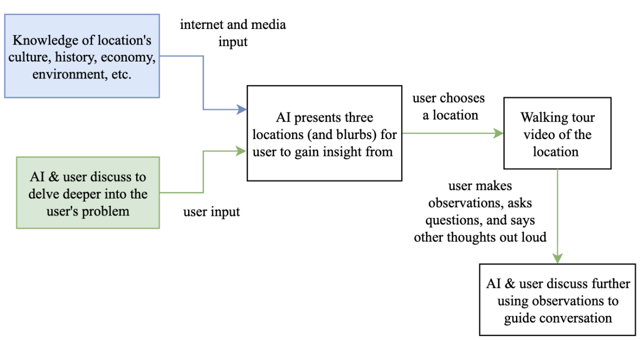
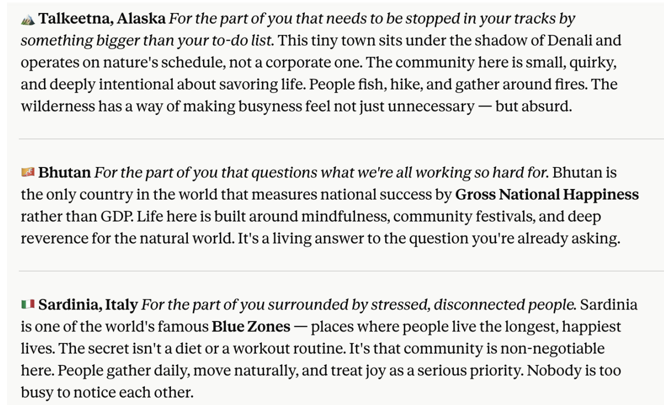


Fig. 2. System Flow Diagram

To begin, the user is prompted to answer "What is something that is bothering you?". The user answers the question and interacts with the AI to identify underlying roots and causes of the problem. For example, the user might respond with "I feel disconnected from nature and my community". Through further conversation, the AI and the user could identify the user's current location of a crowded city as one root cause. When the user feels ready to move on, the user can prompt the AI to retrieve three locations with brief introductory blurbs that will provide new perspectives on the user's problem. The user then chooses one of the three locations presented based on which location they find most interesting from the blurbs (Fig 2).



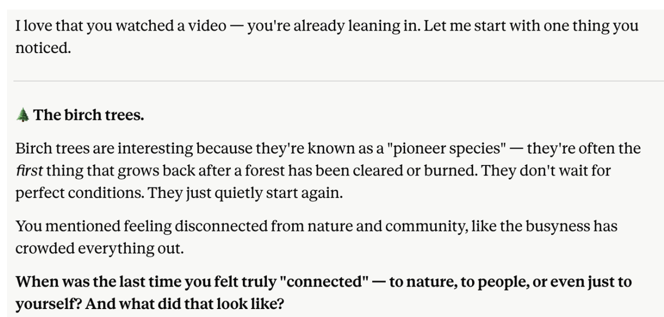
Talkeetna, Alaska *For the part of you that needs to be stopped in your tracks by something bigger than your to-do list.* This tiny town sits under the shadow of Denali and operates on nature's schedule, not a corporate one. The community here is small, quirky, and deeply intentional about savoring life. People fish, hike, and gather around fires. The wilderness has a way of making busyness feel not just unnecessary — but absurd.

Bhutan *For the part of you that questions what we're all working so hard for.* Bhutan is the only country in the world that measures national success by **Gross National Happiness** rather than GDP. Life here is built around mindfulness, community festivals, and deep reverence for the natural world. It's a living answer to the question you're already asking.

Sardinia, Italy *For the part of you surrounded by stressed, disconnected people.* Sardinia is one of the world's famous **Blue Zones** — places where people live the longest, happiest lives. The secret isn't a diet or a workout routine. It's that community is non-negotiable here. People gather daily, move naturally, and treat joy as a serious priority. Nobody is too busy to notice each other.

Fig. 3. AI Selected Locations

After the user chooses a location, the user watches a couple minutes of a walking tour of the location. While watching, the user states anything in the video that piques their interest. For example, if the user picked Talkeetna, Alaska, some items or scenes the user might find interesting are a sign that reads "Denali Brewpub", lots of birch trees, tiny shed-like houses, sparse towns, etc. The user then puts these thoughts into the AI chat, and the AI uses what the user finds interesting to teach the user more about the area as well as provide perspective on their problem (Fig 3).



I love that you watched a video — you're already leaning in. Let me start with one thing you noticed.

The birch trees.

Birch trees are interesting because they're known as a "pioneer species" — they're often the *first* thing that grows back after a forest has been cleared or burned. They don't wait for perfect conditions. They just quietly start again.

You mentioned feeling disconnected from nature and community, like the busyness has crowded everything out.

When was the last time you felt truly "connected" — to nature, to people, or even just to yourself? And what did that look like?

Fig. 4. AI Using User Observations

B. Technical Details

The technical implementation revolves around using a LLM to learn and help a user consider new ways of tackling their problem. Specifically, by providing the LLM with the user's problems and interests as well as directing the LLM to use experiences from around the world, the user is able to learn from others' experiences and gain insight on their own problem(s).

The first challenge is that the AI needs to understand not only the overall challenge the user faces but also the causes behind why the user feels the challenge is insurmountable. To do this, our initial AI prompt tells the AI to "try to figure out the underlying problems and help them think through what their problem(s) is/are". Additionally, we use one-shot prompting by providing a model conversation in the initial prompt. After that, the AI is given continuous user input and feedback throughout the process to craft a personalized experience specific to the user's situation and challenge.

Another challenge we face is figuring out the user's interests to provide an experience they are invested in and feel connected to. As stated in the previous paragraph, the user guides the conversation with their input. Rather than hardcoding a location or things to notice at the location, the user chooses what they are drawn to. Additionally, the AI is told to let the user guide the conversation and use questions to prompt the user instead of telling the user how to feel.

The third challenge is creating an immersive environment that represents the location. Rather than landmarks and stereotypical traits, we wanted to capture a location using people living on a daily basis and staples to their lives. To do so, we use walking tour videos that travel through the area without any additional voiceovers. This lets the user learn more about the area in ways media may not cover and, by having the user explore interactively, we have the user direct their own world learning

IV. Study

We conducted the study on one Northwestern Student. The user was not familiar with the project prior to the study. The study was conducted one-on-one. First, I entered the initial prompt into Claude which included instructions to identify the user's general problem and root underlying causes behind the problem, generate locations relevant to the user's problem, and a model conversation between the user and AI. Then I had the user do the following:

1) Discuss his problem with the AI
2) Pick one of three locations that the AI identified were relevant to the user's problems
3) Watch a couple minutes of a walking tour in the location and say anything he found interesting or any thoughts he had in general out loud
4) Continue the conversation with AI framed by the user's observations

A. Measures

Given the explanatory nature of the prototype and small sample size, I focused on qualitative measures instead of quantitative measures. Specifically, I paid attention to the following:

1) Whether the user was able to learn more about the location he picked
2) Whether the user was able to learn more about their problem and alter his thought process toward the problem
3) How effective the AI guided the user to a new perspective about his problem using what the user learned about the world

1) RQ1: Users learning about the world : The user picked a location that he had never been to in real life but was already very knowledgeable about and had relatives in. Thus, while he was able to learn about the location, most comments while watching the walking tour video focused on the familiarity of the area rather than interesting or surprising things. This reveals that the prototype currently doesn't help users learn more about areas that they are already very familiar with, especially when they already know a lot about the local culture. From this observation, more testing would be required to see whether the AI should never have suggested a location the user already knows very well or whether the representation of the local area was insufficient.

2) RQ2: Users learning more about their problem and how to address it: By interacting with the video and choosing a location the user wants, the AI was able to learn more about the user and what the user cares about. Additionally, the AI used the user's observations and curiosities to direct the conversation which meant the user's interests were integrated in the interaction. During the user's interactions with the world, he paid attention to the surroundings shown in order to discover facts about the local area. This focus immersed the user in the video and thus location. Generalizing, interaction leads to active participation by the user which can reveal facts about the user. By training on these facts, AI can provide more personalized answers that resonate with and interest the user.

3) RQ3: Users gaining new perspective(s) to approach their problem using what they have learned about the world: After the interaction, the user felt a stronger longing to go to the location and felt that going to the location would solve their problem of feeling disconnected to their culture. The location the user picked happened to be where the user's parents are from and thus has many aspects of the culture the user felt disconnected to. While the user was watching the video about the location, he kept noticing he would not fit in, noting both physical and non-physical traits like his looks, accent, and behavior. Instead of helping the user overcome their problem, watching people live their daily lives made the user feel like other people could live proudly with their culture but that the user would be unable to. The LLM did not bridge the gap in the following discussion and did not explore this disconnect felt by the user. Again, this suggests that there is a gap in picking a location for the user to explore, the LLM guiding the user to overcome their problem, or both.

V. Discussion

WorldMirror aims to provide users an immersive experience in another location so that they can learn about how locals live and gain new perspectives to apply to their own life problems. While our one user study showed flaws in our current system when trying to achieve the goals listed, there were some design elements we hope to expand on that did help the user partially achieve the goal.

1) Design lesson 1:Users should be guiding the conversation: Prior to watching the video of the walking tour, the user was able to effectively steer the conversation because the AI prompted the user without adding much input on how the user should think. Instead, the AI provided short reflections of what it learned from the user's query to verify and proposed questions to the user. During this part of the interaction, the AI was able to uncover more about the user's problem and connect it to underlying issues to gain a more holistic view of the user's wants and needs. In contrast, after the interaction, the AI prompts had more statements that extrapolated from the user's thoughts and observations. While the AI still prompted questions to the user, the questions felt like the AI was pushing the user to one answer. Thus, the user had less agency over the conversation which was then reflected by the user's dissatisfaction in the AI's push towards one specific solution.

2) Design lesson 2: Interaction promotes user personalization, immersion, and interest: A second technique is to always work at the level of the full distribution of scores for a given activity and to expose that distribution in multiple coordinated views. When the user scores an activity, the system computes scores for all countries in one batch. From that vector, it generates a choropleth, ranks every country, and extracts the top and bottom countries.

In the study, this combination mattered. Participants did not come in with finely targeted questions like "Compare country X and country Y for this activity." They started with some rough expectation (for example, "I think this should be high in one region"), then scanned the map and rankings. What triggered new questions were the extremes and outliers: a country they never thought about appearing very high, or a country they expected to be high appearing low. Once those contrasts were visible, participants began to ask why.

The generalizable point is that if we want people to ask questions they did not already have, we should show them the whole shape of the model's beliefs, not just respond to a single query. Representations that make extremes and contrasts obvious – like distributions, rankings, and explicit tails – are useful not only for analysis but also as triggers for curiosity.

A. Limitations

This paper has multiple limitations. First, only one user study was conducted. While the user study helped us identify some gaps our current process does not address, one user study is not enough to reveal how a broader population would use the system.

Second, the participant was a Northwestern student who was asked to participate in the study and knew the person conducting the study. Thus, the study does not help identify whether the broader population would want to use the system nor how people with different backgrounds from our one user case would view the system.

Third, the process tested was a low-tech unrefined version of the design. Thus, there were breaks in the flow of the process (ie, between finding videos of walking tours and chatting with the AI), missing features, AI prompts that had only been refined twice, and more that could have changed the effectiveness of the system.

B. Future work

The paper focuses on a rough prototype that implements simple versions of the key design concepts we plan to use in achieving our goal. In the future, some refinements include prompt engineering, other immersive techniques, and accommodating users who want to explore the world even without coming in with a personal problem.

First, on the prompt engineering side, the current process only uses one-shot prompting techniques. For future work, we could explore other prompt engineering techniques like chain-of-thought to force AI reasoning and agent components for self-consistency and breaking the design goals down. Additionally, to increase model effectiveness, limiting possible choices of location for the user would allow for specific refinements in information regarding the limited locations while still leveraging user interests to guide the conversation.

Second, we would like to increase immersion and world learning. One way to do this is to draw from other media focused on the location besides the walking tour video like personal and historical narratives, videos of people doing activities, quotes, news articles, books, etc. Another way to increase immersiveness is to identify portions of the video that are more relevant to the user (ie, if the user feels disconnected from friends and family then show the part of the video that focuses on community building at the location). Additionally, by appealing to senses beyond sight like sound and increasing interaction beyond saying observations aloud, the user can learn about the world and feel more connected to people around the world. The locations in WorldMirror are also limited by the requirement that they have walking tour videos online. On top of that, the quality and effectiveness of walking tour videos vary depending on the popularity of the location and the person recording the video. In future iterations, we hope to use AI in conjunction with google street view and walking tour videos to capture dynamic snippets of people living their lives that effectively assist the user in altering how they view their problem.

Finally, the entry point of the process currently requires the user to answer a problem. However, oftentimes, people can reflect and learn more about their own life through world learning without a clearly defined problem. Thus, in future work, we hope to explore whether changing the entry point of the problem to picking a location would still promote self-reflection and world learning for the user. Additionally, in order to address users who want to learn more about the world, we want the world learning derived from our process to differ from other sources of world learning by focusing on defining the world through peoples' daily experiences and lives.

VI. Conclusion

While existing solutions separate information, immersion, and self-reflection, WorldMirror strives to unify them, by using personal challenge and misalignment as an entry point. Immersion is achieved through the first-person digital model of a chosen location, allowing users to both view and engage with elements in the vicinity. Information is achieved through AI-aided conversation which provides users with context for the elements that pique curiosity. The information is representative, yet curated in an attempt to address the user's initial problem. Self-reflection is embedded throughout as the AI flips the script in having users answer introspective queries designed to help users to revisit their issue as new information is presented. The essence of WorldMirror is that as users learn more about a place, they simultaneously learn more about themselves.

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